

Spatio-Temporal Stochastic Recurrence Model for Extreme Oceanographic Data

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Why model extreme oceanographic events?

Extreme events matter

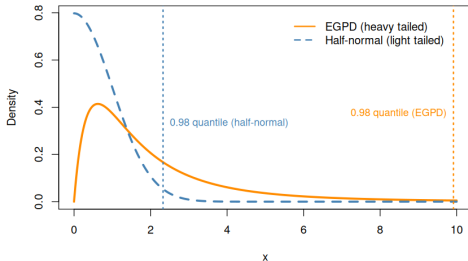
Storms, strong waves and high winds can pose important risks to coastal areas and marine infrastructures.

Oceanographic data



Heavy tails
and
spatio-temporal dependence

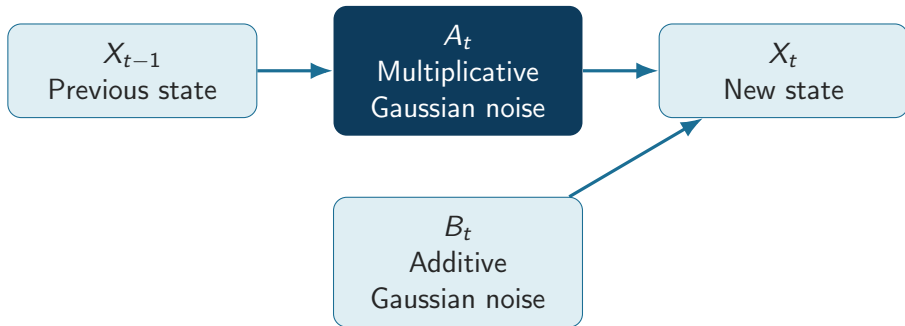
Heavy tail: EGPD vs Normal



Gaussian tail (light tail) $\neq$ Pareto type tail
(heavy tail)

A simple model with an interesting property

$$X_t = A_t X_{t-1} + B_t$$

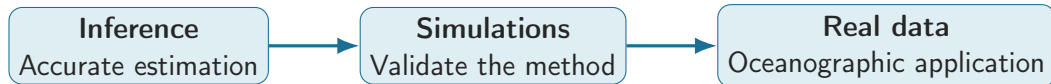


The challenge: inference in high dimension

We need to recover the model parameters
accurately and within a reasonable computation time.

$$4(N_s + 1)$$

parameters



Want to know how?

Come and see me at my poster!